

Table 1. *G36-Degrad* general information

Building	Location	Real/simulated	Control strategy	Number of faults	Number of features	Sampling rate	Data duration
Medium-sized office building	Tuscaloosa, AL, USA	Simulated	G36	12	114	5-min	1 year

Table 2. Details of *G36-Degrad* fouling fault scenarios. Faults were injected during occupied mode duration, which was weekdays (Mon-Fri) from 7 a.m. to 7 p.m.

Fault abbreviation (CSV file name)	Fault type	Fault intensity		Method of fault imposition
		Heat transfer coefficient decrease rate	Pressure drop increase rate	
CooCoiAirFou_he7_pre30	Cooling coil airside fouling	7%/yr	30%/yr	1) Modifying the pressure drop equation from $\dot{m} = k \cdot \sqrt{\Delta p}$ to $\Delta p = coef_f_{dp} \cdot \frac{1}{k^2} \cdot \dot{m}^2$ by introducing a time-dependent pressure drop coefficient ($coef_f_{dp}$). 2) Directly applying fouling effect on heat transfer to heat flux through $\dot{Q}_{condensor,f} = coef_{f_{HT}} \cdot \dot{Q}_{condensor,c}$, where $\dot{Q}_{condensor,f}$ and $\dot{Q}_{condensor,c}$ represent the heat transfer rates of the condenser under the fouled and the clean conditions, respectively.
CooCoiAirFou_he14_pre200		14%/yr	200%/yr	
ChiConWatFou_he4_pre250	Condenser waterside fouling*	4%/30 days	250%/30 days	1) The same pressure drop equation modification as above. 2) Modifying the convection heat transfer rate equation from $\dot{Q} = U_{air} \cdot \Delta T$ to $\dot{Q} = coef_f_{HT} \cdot U_{air} \cdot \Delta T$ by introducing a time-dependent heat transfer degradation coefficient ($coef_f_{HT}$).
ChiConWatFou_he28_pre50		28%/30 days	50%/30 days	

* Faults start on day 170th.

Table 3. Details of *G36-Degrad* non-fouling fault scenarios. Faults were injected during occupied mode duration, which was weekdays (Mon-Fri) from 7 a.m. to 7 p.m.

Fault abbreviation (CSV file name)	Fault type	Fault intensity (starting rate, increase rate)	Method of fault imposition
SupDucLea_st1_end1.4	AHU supply air duct leakage	(1%, 1.4%/yr)	Adding a time-dependent parameter to the AHU air volume model.
SupDucLea_st1_end7		(1%, 7%/yr)	
SupDucLea_st10_end1.4		(10%, 1.4%/yr)	
SupDucLea_st10_end7		(10%, 7%/yr)	
SupDucLea_st25_end1.4		(25%, 1.4%/yr)	
SupDucLea_st25_end7		(25%, 7%/yr)	
TSup_m1	AHU supply air temperature sensor bias	-1 °C/yr	Overwriting the temperature sensor with faulty measurement.
TSup_p1		+1 °C/yr	

Additional information

- Additional helpful document(s):
 - Chu, M., Lu, X., Fu, Y., O'Neill, Z. & Wen, J. in Proceedings of eSim 2022: 12th Conference of IBPSA-Canada (eSim, Vol. 12). IBPSA-Canada, Vancouver, Canada
- All fault simulations span from $t = 0$ sec. (January 1st, the start of the year) to $t = 31,536,000$ sec. (December 31st, the end of the year), covering a full year (365 days) in each CSV file, with one exception: Waterside fouling faults (noted by * in Table 2) were not injected until a certain hot day (day 170th, $t = 14,601,600$ sec.) since the cooling plant was not active until then.
- Please note, since this dataset represents a simulated building, the baseline (fault-free) data was intentionally generated with the same timestamp as the fault data. This was achieved by applying identical outdoor conditions (e.g., outdoor air humidity and temperature), occupancy load, time of day, simulation period, and other related parameters when generating the dataset from the simulated model. This approach allows researchers to compare the corresponding baseline and fault/attack datasets without confounding variations from external factors.
- Fig. 1 shows the schematic of the corresponding AHU system. The system is a single-duct multi-zone VAV AHU connected with five VAV terminal boxes that serve five zones (four exterior and one interior zone) on a single floor.

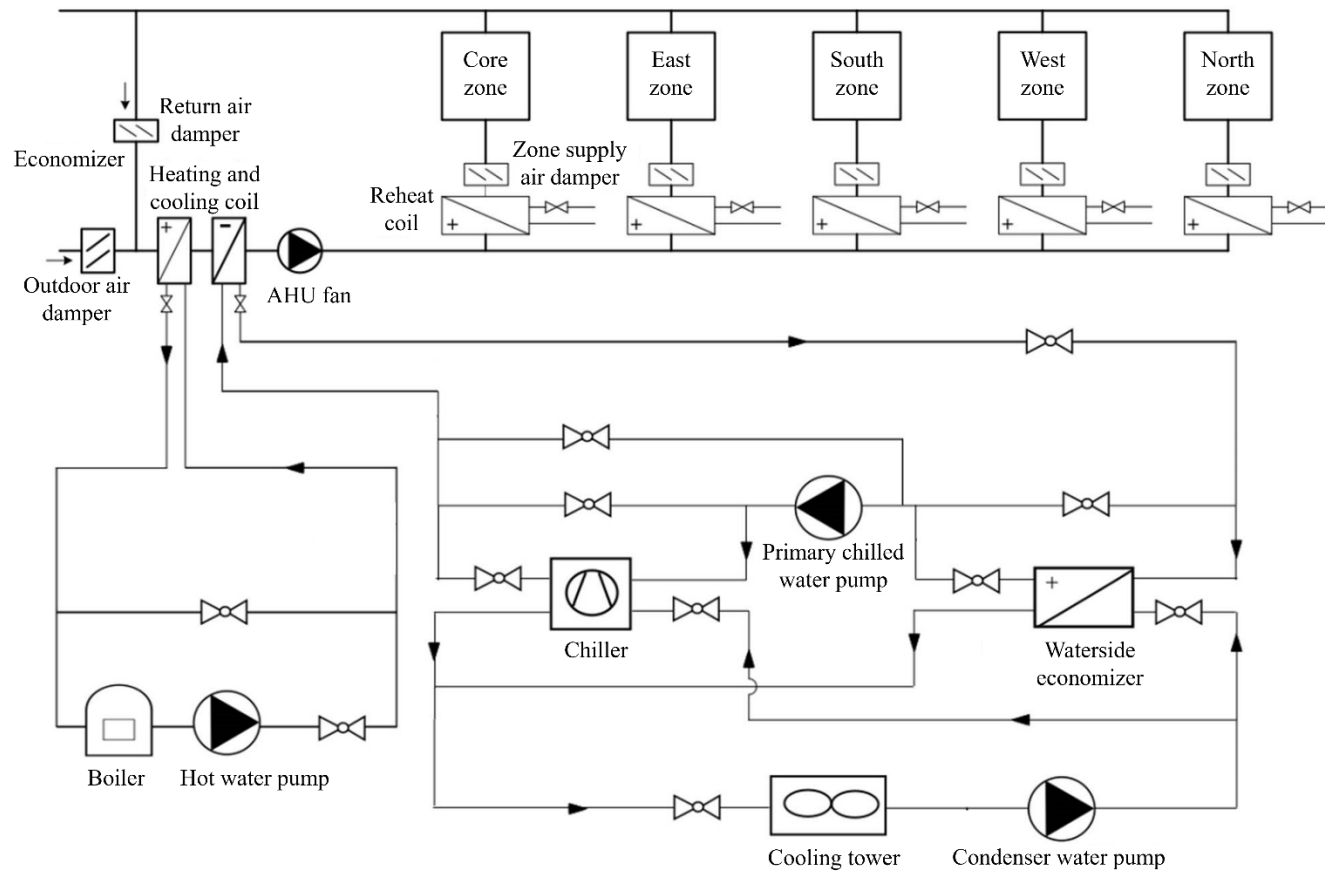


Fig. 1. Schematic of the AHU diagram, used to generate the dataset.